CSE - 641 Computer Vision

Weekly Project Report – 3

Ahmedabad University

Genotype-to-Phenotype Modeling of Alzheimer's Disease Using MRI Radiomics and Polygenic Risk Scores

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***Abstract*—Alzheimer disease is a disorder that affects the brain leading to loss of memory and progressive impairment of the thinking ability. Studies indicate that genetic factors as well as changes in the brain structure are significant in the pathogenesis of Alzheimer disease. However, the correlation between the inheritable genetic risk and the change in the brain structure is not well understood. The presented project seeks to examine the association between genetic vulnerability and brain structure by means of Polygenic Risk Scores (PRS) and MRI-based radiomic characteristics. The features of important brain areas including the hippocampus and the entorhinal cortex, will be extracted using MRI scans of the Alzheimer Disease Neuroimaging Initiative (ADNI) dataset. To determine inherited genetic risk, PRS will be calculated. Machine learning models will be then used to identify the relationship between genetic risk and features of MRI. The aim of the study is to gain more insight into the role of genetic factors in the early brain changes of the Alzheimer disease potentially before symptoms appear.**

I. Introduction

Alzheimer disease is an advanced disorder of the brain which gradually impairs the memory, ability to think and general brain performance. It is connected with when brain tissue is gradually lost throughout time. These changes are usually studied by structural magnetic resonance imaging (MRI), particularly in those parts of the brain that become affected at an early stage in the disease, the hippocampus, entorhinal cortex and temporal lobes [1]. Meanwhile, genome wide association studies of large sizes have enhanced the insights into the genetic etiology of Alzheimer. The genetic risk variants that have been identified through such studies are too many and predispose an individual with the probable cause of the disease [2]. These variants may be used together in a polygenic risk score (PRS) i.e., to sum up inherited genetic risk in a single value [3]. Both MRI markers and PRS can be used to understand the disease, but this is in two dimensions- brain structure and genetic risk.

There have been many research studies that attempted to improve early detection of Alzheimer disease with use of brain imaging methods. Structural magnetic resonance imaging is one of the methods that have been used majorly as it is capable of showing the changes that take place in the brain regions which are affected by disease. One of the affected regions is the hippocampus which is part of the brain that is affected in the Alzheimer disease as it is the part associated with memory. Previous studies were interested in the size of the brain regions. However with the use of the radiomics method, more quantitative features of the MRI images may be obtained that may potentially reveal more details. The authors of the paper [4] would like to create a model of radiomics using MRI images of the hippocampus of the brain to aid in the identification of patients suffering from Alzheimer disease. This would be achieved by the use of data from various cohorts to ensure the reliability of the model. Radiomic features would be achieved by the use of MRI images of the hippocampus. This would then be used to create a model that would aid in the differentiation of patients suffering from Alzheimer disease and healthy patients. This would be achieved by the use of MRI images and the expression of genes present in the blood. This would then attempt to establish a connection between the genetic features and the biological processes aiming to have a better understanding of the changes taking place in the brain as shown by MRI images.

II. Dataset Description

The data used in this project is obtained from the **Alzheimer's Disease Neuroimaging Initiative (ADNI)**, a multi-centered longitudinal large-scale study established to identify and validate biomarkers for the early detection and monitoring of Alzheimer's disease progression.

The study participants are categorized into these primary groups:

- Cognitively Normal (CN)
- Significant Memory Concern (SMC)
- Early Mild Cognitive Impairment (EMCI)
- Late Mild Cognitive Impairment (LMCI)
- Alzheimer's Disease (AD)

ADNI provides multimodal data, including:

- Structural Magnetic Resonance Imaging (MRI)
- Positron Emission Tomography (PET) scans (FDG-PET and Amyloid PET)
- Cerebrospinal Fluid (CSF) biomarker values
- Genetic data (e.g., APOE genotype)
- Neuropsychological assessment scores (e.g., MMSE, CDR, ADAS-Cog)
- Demographic information

The data were collected across multiple phases based on the years it was collected, namely:

- ADNI-1
- ADNI-GO
- ADNI-2
- ADNI-3
- ADNI 4

These phases enable longitudinal analysis of disease progression over time. The dataset is cross-standardized across acquisition sites, ensuring consistency and reliability. Due to its comprehensive and longitudinal nature, the ADNI dataset is widely used in machine learning and deep learning research for classification, progression prediction, and biomarker discovery in Alzheimer's disease studies.

TABLE I: Summary of Selected ADNI Dataset Parameters

Parameter	Selection	Description
ADNI Phases	ADNI-2	These phases differentiate based on the number of years.
Imaging Modality	Structural MRI	Used for extracting structural brain features and radiomic biomarkers.
MRI Sequence	T1-weighted MPRAGE	Standard structural sequence widely used for brain morphology studies.
Field Strength	3 Tesla (3T)	Provides higher spatial resolution and better signal-to-noise ratio.
Visit Type	Baseline and Screening	Ensures each subject contributes a single scan, avoiding longitudinal duplication.
Subject Groups	CN, EMCI, LMCI, AD	Represents healthy controls, early impairment, and Alzheimer's disease patients.
Image Format	NIfTI (.nii)	Standard neuroimaging format compatible with most analysis pipelines.
Genetic Data	APOE Genotype	Known genetic risk factor strongly associated with Alzheimer's disease.

A. Dataset Selection

In this research, the entire data was then narrowed down to one subset to guarantee uniformity and radiomics analysis. Data contains structural MRI scans collected with the T1-weighted MPRAGE sequence, as part of the ADNI-2 and ADNI-3 phases owing to the high spatial resolution and standardized experimental procedures. Scans taken at 3 Tesla (3T) were used exclusively because the scans with high field strength offer better ratio of signal to noise and a better image as compared to lower field strength scans which are used in quantitative extraction of features. Baseline visits were chosen to ensure one scan per subject and avoid longitudinal bias.

III. Literature Review

A. Longitudinal structural MRI-based deep learning and radiomics features for predicting Alzheimer's disease progression

The authors in this paper show how a longitudinal MRI-based paradigm can be utiliized to forecast the subjects who will progress to Alzheimer disease from mild cognitive impairment (MCI). They build a two-stage deep learning model on the T1-weighted images of the Alzheimer Disease Neuroimaging Initiative dataset. They first train a 3D ResNet18 model on individual visit MRI volumes to enable it to learn spatial brain features related to disease risk with a survival-based ranking loss. The learned feature representations are then used as input to a time-varying LSTM that trains to learn multiple MRI scans over time, accounting for irregular time intervals between visits. The attention mechanism identifies the most relevant time points. They also discard gray matter radiomics features like shape, intensity, and texture features, and discard them using PCA, and then combine them with deep features. The final model is being placed within a survival analysis framework to estimate individualized risk of time-to-conversion, rather than just binary classification.

B. Predicting Conversion from Mild Cognitive Impairment to Alzheimer's Disease Using a Vision Transformer and Hippocampal MRI Slices

The deep learning models are employed to predict the development of Mild Cognitive Impairment (MCI) to Alzheimer's Disease (AD) from MRI scans. Most of the existing studies were conducted using Convolutional Neural Networks (CNNs) because they possess the capability to learn spatial information from images of the brain. Some studies have been demonstrated to possess high accuracy, but there were problems such as data leakage, where data from the same patient was used for training and testing, which could potentially lead to biased results.

Recently, Vision Transformers (ViTs) have been proposed for image processing. Although Vision Transformers have been demonstrated to possess high accuracy for Alzheimer's classification, few studies have been conducted to explore their application for predicting MCI to AD conversion.

This paper continues from the existing studies by employing a Vision Transformer model that was pre-trained on hippocampal MRI slices. It strictly separates data for each subject and compares the results to a 3D CNN model. The results indicate that Vision Transformers possess the capability to produce similar results to CNN models.In [5], genetic markers as single nucleotide polymorphisms (SNP) are compared to brain imaging measurements carried out by the use of MRI scans to perform this analysis. A statistical test is then used to find significant correlations between genetic variants and specific neuroimaging features regarding the Alzheimer disease.

Polygenic Risk Scores provide an understanding of how combination of genes of a person may influence chance of developing Alzheimer's disease. The authors of the [6] explored many case control and by proxy studies data through a genome wide association method to determine genetic differences associated with Alzheimer's disease. The research shows that when PRS used in combination with APOE carrier status could stratify individuals on the risk and the median age at onset of high-risk individuals differs. This paper brings

Fig. 1: Pipeline for MRI data selection and preprocessing from the ADNI dataset

attention to the fact that it is better to combine the numerous genetic risk factors into one score that would further predict the disease susceptibility. Though the study mainly concentrates on the genetic side it also forms the basis of the association of genetic risk with alteration of the brain which show that early change in the structure of the brain could occur in people with high PRS. In the context of our project this provides the reasoning behind the integration of PRS and MRI-based radiomic features in the effort to investigate the development of cumulative genetic risk into subtle phenotypes in hippocampal and entorhinal cortex structures which would possibly support the earlier detection of the advancement of Alzheimer disease.

IV. Methodology

A. MRI Data Acquisition and Preprocessing

Structural MRI scans were obtained from the Alzheimer's Disease Neuroimaging Initiative (ADNI) database. For this study, T1-weighted three-dimensional magnetization-prepared rapid acquisition gradient echo (3D MPRAGE) images from the ADNI3 phase were used. Only scans acquired during the screening visit on 3-Tesla scanners were included in order to maintain consistency in image quality and reduce variability associated with different field strengths. ADNI employs standardized imaging protocols across participating sites and performs centralized quality control to ensure that images meet predefined standards and are free from major motion or acquisition artifacts.

After downloading the data, the MRI scans were converted from DICOM format to NIfTI format and organized into a structured dataset to support automated processing. Several preprocessing steps were then applied to improve image consistency across subjects. First, N4 bias field correction was performed to reduce intensity inhomogeneity caused by scanner-related variations. Next, skull stripping was applied to remove non-brain tissues such as the skull and surrounding structures, leaving only the brain for further analysis. The resulting brain images were then resampled to an isotropic resolution of $1 \times 1 \times 1$ mm³ to standardize voxel size across all scans.

Finally, voxel intensities were normalized using z-score normalization to reduce differences in intensity distributions between subjects and scanning sites. Together, these preprocessing steps ensured that the MRI images were standardized in terms of spatial resolution and intensity characteristics, providing reliable inputs for subsequent segmentation and radiomic feature extraction.

V. Work Completed in Week-3

We completed the following work during this week:

- We loaded the MRI data and changed the DICOM images into NIfTI files and arranged the data in a structured format to process it automatically.
- The N4 bias field correction was used to correct for inhomogeneity of intensity, and skull stripping was performed to get rid of tissue other than the brain.
- The images were resampled to $1 \times 1 \times 1$ millimeter isotropic resolution, and we applied a z-score

intensity normalization to make voxel intensity the same.

- We prepared the preprocessed data to be segmented and undergo radiomic analysis.

VI. Plan for Week-4

For the upcoming Week 4, we aim to segment the preprocessed brain images, extract and store the radiomic features, conduct EDA, and develop machine learning models for Alzheimer's classification or prediction.

- Carry out logical segmentation of the brain tissue from the images received after preprocessing.
- Determine those brain regions that hold significance for the analysis of Alzheimer's.
- Access radiomic features (texture, intensity, and shape) from the segments delineated.
- Create a database of these extracted features for use in developing a machine learning model.
- Conduct EDA to determine the distribution of each feature and find out how the features correlate with each other.
- Start developing the machine learning models for classification or prediction tasks related to Alzheimer's disease.

References

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